Final value

1 $\$3000$ is invested at an interest rate of 5% p.a. compounded annually. Find the value of the investment after 17 years.

2 $\$3000$ is invested at an interest rate of 4% p.a. compounded quarterly. Find the value of the investment after 17 years.

3 $\$2000$ is borrowed at an interest rate of 12% p.a. compounded monthly. Find how much is owed at the end of 2 years.

4 $\$10\,000$ is invested at an interest rate of 2.7% p.a. compounded monthly. Find how much the investment is worth after 18 months.

5 $\$13\,000$ is borrowed at an interest rate of 2.5% p.a. compounded semi-annually. Find how much is owed after 3.5 years.

6 A deposit of $\$6000$ attracts interest at a rate of 2.5% p.a. compounded monthly. Find how much is in the account after 3.5 years.

7 Ryan borrows $\$4000$ at a rate of 5.5% p.a. compounded weekly. Assuming he makes no repayments, find how much he owes after 1 year . Assume there are 52 weeks in a year.

8 $\$5000$ is borrowed at an interest rate of 3.5% p.a. compounded daily. Find how much is owed after 4 years. Assume there are 365 days in a year.

Interest

9 $\$2000$ is invested at an interest rate of 4% p.a. compounded quarterly.

a Find the value of the investment after 15 years.

b Find the amount of interest accrued over the 15 years.

10 $\$5000$ is borrowed at an interest rate of 12% p.a. compounded monthly.

a Find the amount owed after 3 years.

b Find the amount of interest accrued over the 3 years.

11 $\$10\,000$ is invested at an interest rate of 2.5% p.a. compounded monthly. Find the amount of interest that is added to the account over 24 months.

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12 $\$11\,000$ is borrowed at an interest rate of 3.5% p.a. compounded semi-annually. Find the amount of interest that is added to the account over 30 months.

13 Nadia borrows $\$12\,000$ at an interest rate of 3.5% p.a. compounded weekly. If she makes no repayments, find the amount of interest that is owed after 3 years. Assume there are 52 weeks in a year.

14 $\$900$ is borrowed at an interest rate of 10.5% p.a. compounded monthly. Find the amount of interest that is owed after 1.5 years.

15 Luke borrows $\$700$ at an interest rate of 17% p.a. compounded daily. Find the amount of interest that is owed after two weeks. Assume there are 365 days in a year.

16 $\$24\,000$ is deposited in a savings account which attracts interest at 1.3% p.a. compounded daily. Find the amount of interest that is accrued over 9 years. Assume there are 365 days in a year.

17 $\$12\,000$ is invested at an interest rate of 0.4% per month compounded monthly. Find the amount of interest that is accrued over 4.5 years .

18 Harry borrows $\$3300$ at a monthly rate of 1.2% compounded daily. He takes out the loan at the beginning of the month of October, and repays it at the end of the month in one lump sum. Find the amount of interest that is paid.

19 Katrina has her property investment currently valued at $\$750\,000$ and she knows that its value increased at a rate of 10% p.a. compounded annually for the last 20 years. Find the value of her investment at the beginning of the 20 years.

Principal, interest rate and time period

20 A $\$2000$ investment grows to $\$4000$ over 2 years accumulating interest which is being compounded monthly. Calculate the interest rate per annum which is being applied. Give your answer as a percentage to two decimal places.

21 What interest rate per annum would be required to double an investment of $\$1000$ in 2 years if the interest is compounded daily? Assume there are 365 days in a year. Give your answer as a percentage to two decimal places.

22 $\$1000$ is placed in a term deposit with a rate of 5% p.a. compounded annually. How many whole years will it take for the balance to increase in value to at least $\$1500$?

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23 Vanessa borrows $\$25000$ at an interest rate of 4.3% p.a. compounded annually in order to pay for her university degree. If she continues into postgraduate study and therefore makes no repayments, how many whole years will it take for her debt to reach at least $\$40\,000$?

24 Nadia invests $\$1200$ in a term deposit with a rate of 2.3% p.a. compounded monthly. How many whole months will it take for the investment to increase in value to at least $\$1500$?

25 Ryan borrows $\$800$ from the bank at a rate of 0.2% per week, compounded weekly. He is unable to make any repayments. Find out how many whole weeks it will take for the amount he owes to increase to at least $\$1300$.

26 Neil invests $\$900$ in a term deposit with a rate of 2.3% p.a. compounded daily. How many years will it take for the investment to at least double in value? Assume there are 365 days in a year.

27 What interest rate per annum would be required to double an investment of $\$P$ in 4 years if the interest is compounded quarterly? Give your answer as a percentage to two decimal places.

28 How many whole years will it take an investment of $\$P$ to at least double in value if interest at 4.2% p.a. is compounded quarterly?

29 Stephen borrows $\$12\,500$ to buy a car. The bank offers a reducing balance loan with interest rate of 4.6% p.a. compounded monthly. Stephen opts to make weekly payments of $\$100$ in order to pay off the loan. Use the financial application on your calculator to answer the following questions. Assume there are 52 weeks in a year.

a What is the balance of the loan after 20 weeks,?

b How long does it take him to pay off the loan in years? Round your answer to two decimal places.

30 Valerie borrows $\$345\,000$ to buy an apartment. The bank offers a reducing balance loan with an interest rate of 2.35% p.a. compounded monthly. Valerie opts to make fortnightly payments of $\$1250$ in order to pay off the loan. Use the financial application on your calculator to answer the following questions. Assume there are 26 fortnights in a year.

a What is the balance, after 100 weeks?

b Approximate how long it takes her to pay off the loan in years. Round your answer to two decimal places.

31 A savings account has a balance of $\$25\,500$ after 10 years of interest at 1.6% p.a. compounded quarterly. Find the initial deposit.

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32 Harry expects to receive a Christmas bonus of $\$7000$ in 6 months time. His credit card incurs interest at 14% p.a. compounded weekly with no interest free period. What is the most he can spend now using his credit card and still be able to pay it off using his Christmas bonus? Assume there are 52 weeks in a year.

33 Jack is aiming to save $\$35000$ after 10 years of investing at an interest rate of 0.1% per week compounded monthly. Find the amount he needs to deposit as a principal. Assume there are 52 weeks in a year.

Financial solvers

34 Aaron borrows $\$15000$ to buy a car. He is charged 6.8% reducible interest compounded monthly. Use the financial solver on your CAS calculator to determine what his monthly repayments will be if he wishes to pay off the loan in 3 years by answering the following:

a Which variable on the CAS calculator do we want to solve for?

b Complete the following table:

c Hence, state the minimum value of his monthly repayments.

Variable	Value
N	
$I\%$	
PV	
FV	
PpY	
CpY	

35 Mr and Mrs Gwen held a mortgage for 25 years. Over that time they made monthly repayments of $\$4500$ and were charge a fixed interest rate of 4.4% per annum, compounded monthly.

a Which variable on the CAS calculator do we want to solve for?

b Fill in the value for each of the following:

c Hence, state how much Mr. and Mrs. Gwen initially borrowed, correct to the nearest dollar.

Variable	Value
N	
$I\%$	
Pmt	
FV	
PpY	
CpY	

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36 Derek borrows $\$50\,000$ at a rate of 9% (per annum) reducible interest compounded quarterly. At the end of each quarter he makes a repayment of $\$1800$.

a Which variable on the CAS calculator do we want to solve for?

b Complete the following table:

c Hence, state after how many quarters the loan will be repaid.

Variable	Value
$I\%$	
PV	
Pmt	
FV	
PpY	
CpY	

37 Xanthe borrows $\$32\,000$ at a rate of 6.5% (per annum) reducible interest compounded monthly. At the end of each month she makes a repayment of $\$380$.

a Which variable on the CAS calculator do we want to solve for?

b Complete the following table:

c Hence, state after how many months the loan will be repaid.

d Calculate the total amount Xanthe paid.

e Calculate how much Xanthe paid in interest.

Variable	Value
$I\%$	
PV	
Pmt	
FV	
PpY	
CpY	

38 Tara borrows $\$15\,000$ and is charged quarterly reducible interest at a rate of 7% per annum compounded quarterly. Use the financial solver on your CAS calculator to determine what her quarterly repayments should be if she wishes to pay off the loan in 7 years.

a Which variable on the CAS calculator do we want to solve for?

b Complete the following table:

c Hence, state the minimum value of her quarterly repayments.

d Calculate the total amount Tara paid.

e Calculate how much Tara paid in interest.

Variable	Value
N	
$I\%$	
PV	
FV	
PpY	
CpY	

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39 Tom has a mortgage of $\$500\,000$ reducible monthly with an annual interest rate of 4.2%. He makes fortnightly repayments of $\$1500$. Assume for this question that there are 26 fortnights in a year.

a Use the financial solver on our CAS calculator to determine how long it takes to pay off the loan. Fill in the value for each of the following, typing an X next to the variable we wish to solve for.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

b Hence state the number of full years it will take to pay off the loan.

c If Tom would like to pay off his loan in 10 years, what fortnightly repayment does he need to make? We can use our CAS calculator to calculate this amount. Fill in the values to be entered into your calculator, typing an X next to the variable we wish to solve for.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

d Hence state the fortnightly repayment to the nearest dollar. Give your answer as a positive number.

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40 Pauline and Brad borrow $\$330\,000$ for a house extension. The bank offers them 2.75% p.a compounded daily. They also make a payment of $\$67$ per day. Assume there are 365 days in a year.

a If N is the number of payments, complete the table of values showing the variables required to use the financial application of your calculator to determine how long it takes for Pauline and Brad to complete their payments.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

b Determine the whole number of years it will take until Pauline and Brad pays back the $\$330\,000$ loan required for the extension.

c If they change their payment to $\$75$ per day, how many whole years will it take until they pay the loan?

d How many years do they save by increasing their payment to $\$75$?

41 Caitlin borrows $\$250000$ to buy a unit and wants to pay it back before she is 33. She just turned 23 years old and the current interest rate is 3.2% p.a. compounded monthly. Emma wishes to find the monthly payment needed to reach her goal.

a If N is the number of payments, complete the table of values showing the variables required to use the financial application of your calculator to determine the monthly payment.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

b Determine the monthly payment required for Caitlin to repay the $\$250\,000$ loan by the time she is 33 years old.

c How much does she pay in total over the term of the loan, ignoring bank fees and other charges?

d If she extends the term of the loan to 25 years, how much will her monthly payment be, assuming interest rates stay the same?

e How much does she pay in total over the 25 years, ignoring bank fees and other charges?

f Hence, determine much money she saves if she pays the loan back after 10 years instead of25years.

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42 Pauline and Jenny take out a loan of $\$120000$ for a new caravan. The loan rate is 3.015% p.a compounded weekly. They decide to make payments of $\$100$ per week.
Assume there are 52 weeks in a year.

a If N is the number of payments, complete the table of values showing the variables required to use the financial application of your calculator to determine how long it takes for Pauline and Jenny to complete their payments.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

b Determine the whole number of weeks it will take until Pauline and Jenny pays back the $\$120\,000$ loan they borrowed for the caravan.

c Assuming all payments are equal in size, how much do Pauline and Jenny pay for the caravan?

d If they decide to triple their payment to $\$300$ per week, how many whole weeks will it take them until they pay the loan?

e Therefore calculate how much money they save if they triple their payment.

43 A student borrows $\$60000$ to pay their student loan. The bank offers a reducing balance loan and charges a student rate of 0.95% p.a. compounded weekly. She wants to pay the loan off completely in 8 years in equal monthly payments. Assume there are 52 weeks in a year.

a If N is the number of payments, complete the table of values showing the variables required to use the financial application of your calculator to determine what monthly payment the student will need to make.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

b State the monthly payment required.

c How much should she pay each month if she wants to pay the loan off in half the time?

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44 A young couple wish to borrow $\$210\,000$. Bank 1 is advertising a reducing balance loan with an interest rate of 3.05% p.a. compounded monthly and quarterly payments of $\$5000$. Bank 2 offers them the deal that they can pay the loan in 60 installments of $\$4500$.

a If N is the number of payments, complete the table of values showing the variables required to use the financial application of your calculator to calculate the number of payments required to pay the loan option for Bank 1.

Variable	Value
N	
$I\%$	
PV	
Pmt	
FV	
P/Y	
C/Y	

b Calculate the number of whole quarters it takes until the loan is paid using Bank 1.

c Calculate the total amount paid to Bank 2 over the duration of the loan.

d Which bank should the couple choose?

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